

FINAL PROJECT

EMPLOYEE ATTRITION ANALYSIS USING PYTHON

1. Project Overview

This project focuses on analyzing employee attrition using Python to identify factors associated with employee turnover. The project involves data preprocessing, exploratory data analysis (EDA), multivariate analysis, and data visualization using Pandas, NumPy, Matplotlib, and Seaborn. The objective is to discover meaningful patterns related to employee attrition that can support better human resource decision-making.

2. Tools Used

- Pandas – Data loading, cleaning, manipulation, and analysis
- NumPy – Numerical computations
- Matplotlib – Basic data visualization
- Seaborn – Statistical and advanced visualizations
- Jupyter Notebook – Project development environment

3. Dataset

- **Source**

IBM HR Employee Attrition Dataset

- **Description**

The dataset contains employee information with 1470 records and 35 attributes.

The important columns include:

- Employee Number
- Age

- Department
- Gender
- Job Role
- Monthly Income
- Job Level
- Total Working Years
- Years at Company
- OverTime
- Attrition (Target Variable)

4. Steps Followed

1. Imported the dataset using Pandas.
2. Performed initial data exploration by:
 - Viewing the dataset
 - Checking dataset dimensions
 - Inspecting data types
 - Generating descriptive statistics
3. Performed data preprocessing by:
 - Checking missing values
 - Checking duplicate records
 - Verifying data types
 - Removing constant columns
 - Analyzing unique values
4. Conducted Exploratory Data Analysis (EDA):
 - Univariate Analysis
 - Bivariate Analysis
 - Multivariate Analysis
5. Created visualizations using Matplotlib and Seaborn, including:
 - Count plots
 - Box plots
 - Correlation Heatmap
6. Interpreted the analytical results and extracted meaningful business insights.

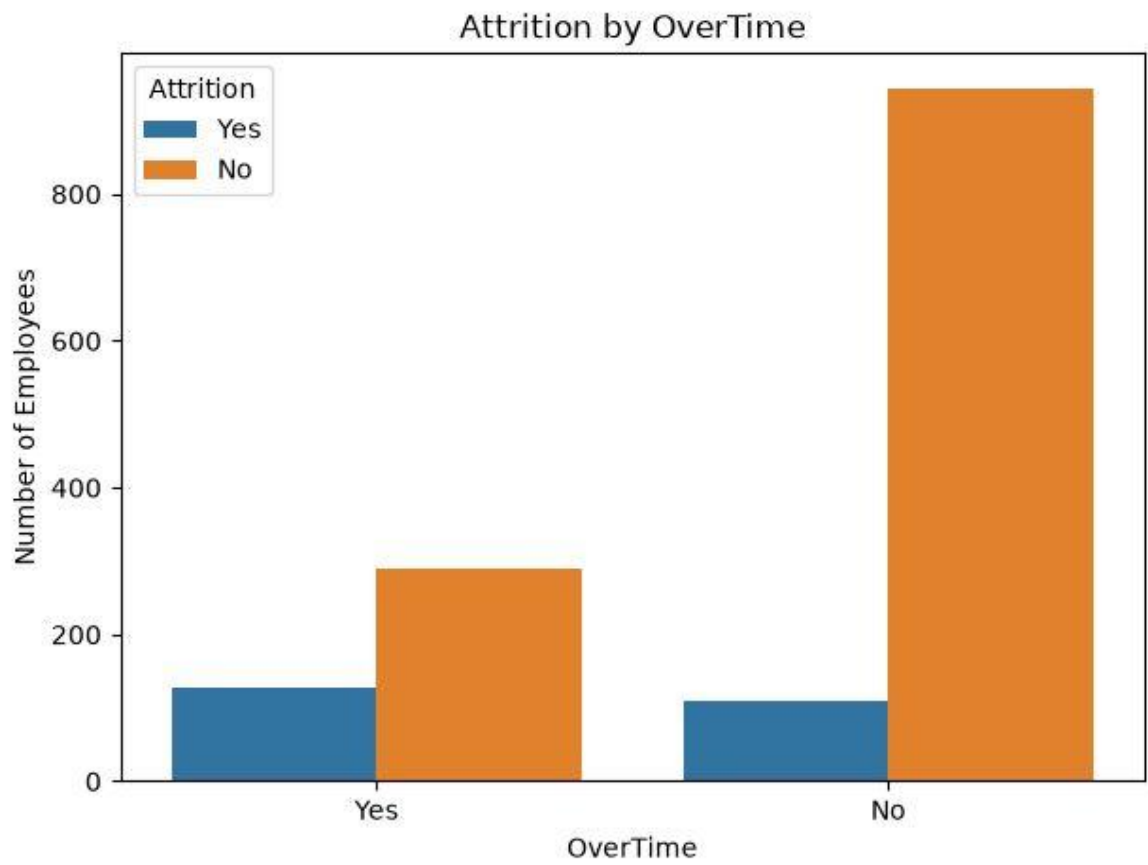
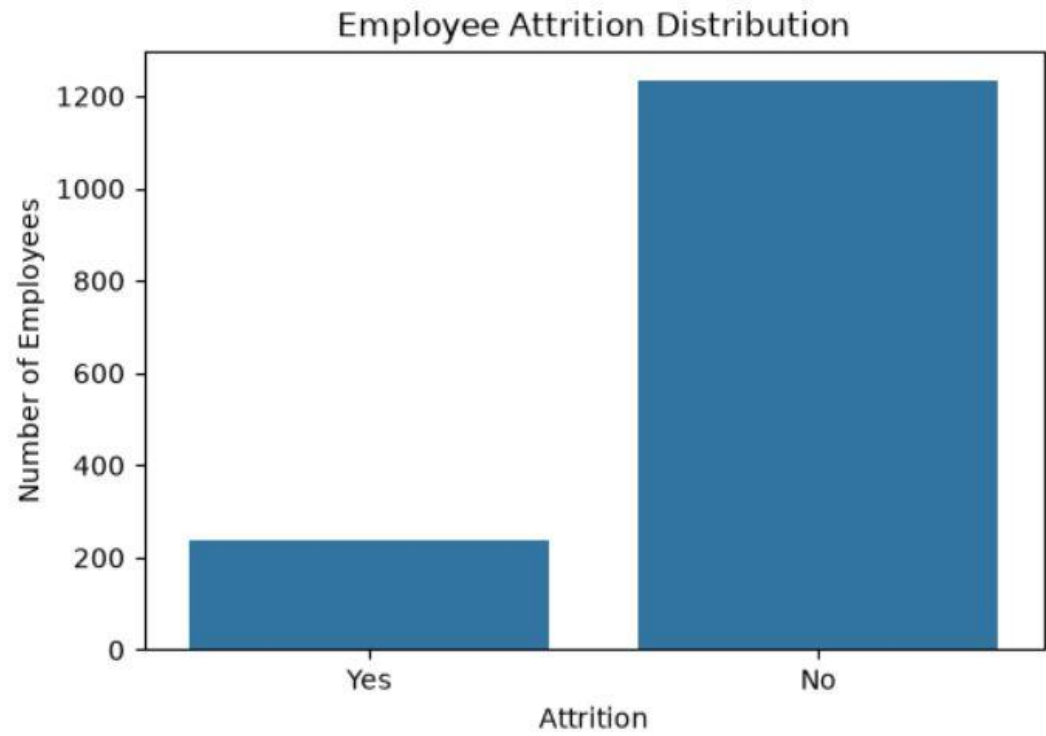
5. Key Insights

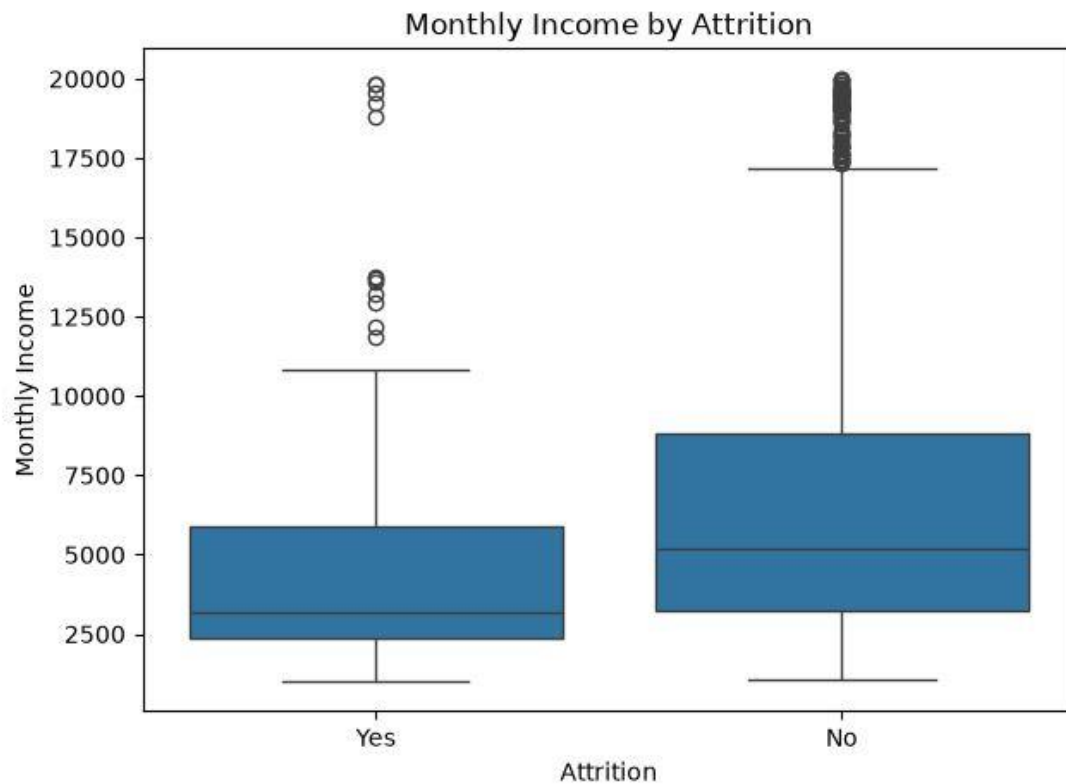
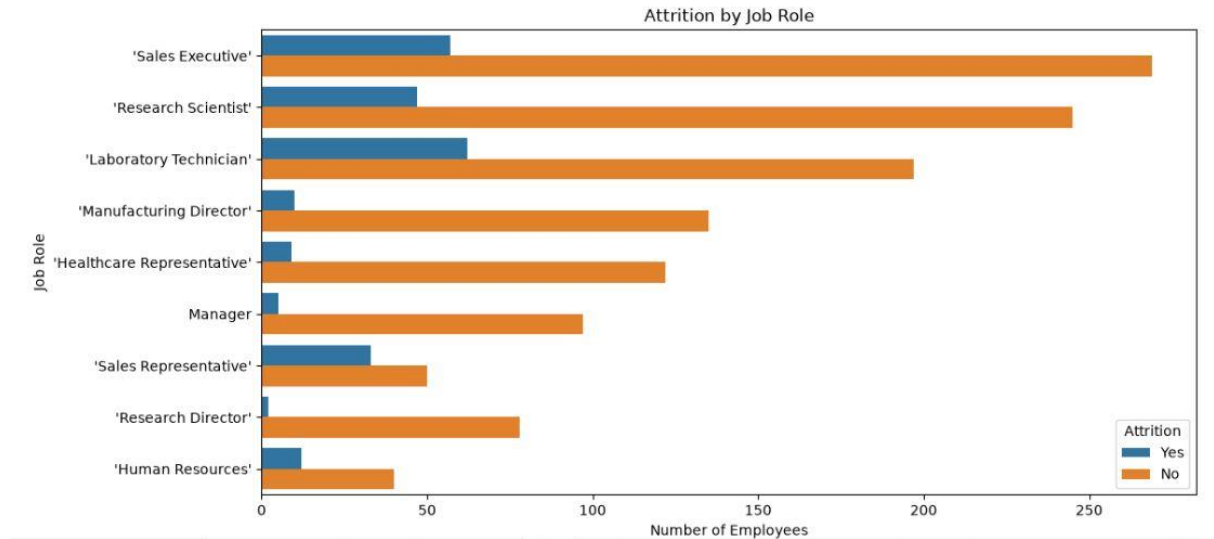
- Most employees remained with the organization, while a smaller proportion left.
- Employees working overtime showed higher attrition than employees who did not work overtime.
- The Sales department experienced the highest employee attrition.
- Sales Representatives recorded the highest attrition among all job roles.
- Employees with lower monthly income and shorter organizational tenure were more likely to leave the company.
- Job Level and Monthly Income exhibited a strong positive correlation.
- Employees with greater work experience generally occupied higher job levels and earned higher salaries.

6. Visualizations

The project includes the following visualizations:

- Employee Attrition Distribution
- Department-wise Employee Distribution
- Job Role Distribution
- Overtime Distribution
- Gender Distribution
- Attrition vs OverTime
- Attrition vs Department
- Attrition vs Job Role
- Monthly Income by Attrition (Box Plot)
- Years at Company by Attrition (Box Plot)
- Correlation Heatmap





7. Files Included

- IBM_HR_Analytics_Employee_Attrition.csv – Dataset
- Employee_Attrition_Analysis_FINAL_PROJECT.ipynb– Jupyter Notebook containing preprocessing, analysis, and visualizations
- Project_Report.pdf – Project documentation

8. How to Use

1. Open the Jupyter Notebook using Jupyter Notebook or JupyterLab.
2. Run each notebook cell sequentially.
3. Review the preprocessing steps, EDA, multivariate analysis, and visualizations.
4. Explore the generated insights and modify the notebook for further analysis if required.

9. Conclusion

This project demonstrates how Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn can be effectively used to analyze employee attrition data. Through data preprocessing, exploratory data analysis, multivariate analysis, and visualization, several factors associated with employee attrition were identified. The insights obtained from this project can help organizations understand workforce trends, improve employee retention strategies, and support data-driven human resource decision-making.